

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279121

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

ZHANG; Jiarui

SEMICONDUCTOR CHIP, SEMICONDUCTOR STRUCTURE, AND SEMICONDUCTOR DEVICE

Abstract

A semiconductor chip includes: a first ID transmission path, penetrating through a substrate of the semiconductor chip in a direction perpendicular to an active surface of the semiconductor chip; a second ID transmission channel, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, and symmetrically distributed with the first ID transmission path based on a central axis of the semiconductor chip, where the central axis passes through a center of the semiconductor chip and is parallel to the active surface; and an ID decoding circuit, coupled to the first ID transmission path or the second ID transmission path, where the ID decoding circuit is configured to receive an ID signal and decode the ID signal to generate a data selection signal, where the semiconductor chip outputs data signals transmitted in a plurality of data transmission paths to corresponding internal receiving circuits respectively.

Inventors: ZHANG; Jiarui (Hefei City, CN)

Applicant: CXMT Corporation (Hefei City, CN)

Family ID: 93645715

Assignee: CXMT Corporation (Hefei City, CN)

Appl. No.: 19/211229

Filed: May 18, 2025

Foreign Application Priority Data

CN 202310621163.0

May. 26, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2024/091608 20240508 PENDING child US 19211229

Publication Classification

Int. Cl.: **G11C5/06** (20060101); **H01L25/065** (20230101); **H01L25/18** (20230101); **H10B80/00** (20230101)

U.S. Cl.:

CPC **G11C5/063** (20130101); **H01L25/0657** (20130101); **H01L25/18** (20130101); **H10B80/00** (20230201); H01L2225/06513 (20130101); H01L2225/06541 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation application of International Application No. PCT/CN2024/091608 filed on May 8, 2024, which claims priority to Chinese Patent Application No. 202310621163.0 filed on May 26, 2023. The disclosures of the above-referenced applications are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] As the demand for electronic products moves towards miniaturization and multifunctionality, packaging technology is also developing towards high density and high integration. For example, at least two memory chips (dies) are three-dimensionally stacked and packaged in a direction perpendicular to a horizontal plane.

[0003] However, the three-dimensional stack structure still faces many technical problems to be solved, for example, how to ensure that data is accurately transmitted to each chip of the three-dimensional stack structure.

SUMMARY

[0004] Embodiments of the present disclosure relate to the technical field of semiconductors, and in particular to a semiconductor chip, a semiconductor structure, and a semiconductor device.

[0005] According to a first aspect of embodiments of the present disclosure, provided is a semiconductor chip. The semiconductor chip includes: a first identification (ID) transmission path, penetrating through a substrate of the semiconductor chip in a direction perpendicular to an active surface of the semiconductor chip; a second ID transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, and symmetrically distributed with the first ID transmission path based on a central axis of the semiconductor chip, where the central axis passes through a center of the semiconductor chip and is parallel to the active surface; and an ID decoding circuit, coupled to the first ID transmission path or the second ID transmission path, where the ID decoding circuit is configured to receive an ID signal and decode the ID signal to generate a data selection signal, where the semiconductor chip outputs data signals transmitted in a plurality of data transmission paths to corresponding internal receiving circuits respectively according to the data selection signal.

[0006] In some embodiments, each of the internal receiving circuits includes: a first data receiving circuit and a second data receiving circuit, where the first data receiving circuit is coupled to a first data selecting circuit, and the second data receiving circuit is coupled to a second data selecting circuit; and the plurality of data transmission paths include: a first data transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, where the first data transmission path is configured to transmit a first data signal; a second data transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, and symmetrically distributed with the first data transmission path based on the central axis, where the second data transmission path is

configured to transmit a second data signal, and the second data signal is different from the first data signal; the first data selecting circuit, separately coupled to the ID decoding circuit, the first data transmission path, and the second data transmission path, where the first data selecting circuit is configured to output one of the first data signal or the second data signal to the first data receiving circuit according to the data selection signal; and the second data selecting circuit, separately coupled to the ID decoding circuit, the first data transmission path, and the second data transmission path, where the second data selecting circuit is configured to output the other of the first data signal or the second data signal to the second data receiving circuit according to the data selection signal.

[0007] In some embodiments, the first ID transmission path is configured to transmit a first ID signal; and the second ID transmission path is configured to transmit a second ID signal, where the second ID signal is different from the first ID signal.

[0008] In some embodiments, the first ID transmission path includes: a first connection pad located on an inactive face of the semiconductor chip, a first path located in the substrate, and a first interconnection structure located on the active surface of the semiconductor chip, where the first connection pad, the first path, and the first interconnection structure are sequentially connected; and the second ID transmission path includes: a second connection pad located on the inactive face of the semiconductor chip, a second path located in the substrate, and a second interconnection structure located on the active surface of the semiconductor chip, where the second connection pad, the second path, and the second interconnection structure are sequentially connected.

[0009] In some embodiments, the ID decoding circuit is coupled to the first interconnection structure or the second interconnection structure.

[0010] According to a second aspect of the embodiments of the present disclosure, provided is a semiconductor structure. The semiconductor structure includes: a logic chip; and a stack structure, stacked on the logic chip, and including: 1st to 2N-th semiconductor chips stacked sequentially, where the 1st to 2N-th semiconductor chips are same memory chips; an inactive face of the 1st semiconductor chip is bonded to the logic chip, and an active surface of a 2i-th semiconductor chip is bonded to an active surface of a 2i-1-th semiconductor chip; and N is a positive integer, and i is a positive integer less than or equal to N; and where each of the semiconductor chips includes: a first ID transmission path, a second ID transmission path, and an ID decoding circuit; the first ID transmission path and the second ID transmission path penetrate through a substrate of one of the semiconductor chips in a direction perpendicular to an active surface of the corresponding one of the semiconductor chips and are symmetrically distributed based on a central axis of the corresponding one of the semiconductor chips, and the central axis passes through a center of the corresponding one of the semiconductor chips and is parallel to the active surface; and the ID decoding circuit is coupled to the first ID transmission path or the second ID transmission path, the ID decoding circuit is configured to receive an ID signal, and the ID signal indicates that one of the semiconductor chips is the 2i-th semiconductor chip or the 2i-1-th semiconductor chip in the stack structure, where the first ID transmission path of the 2i-1-th semiconductor chip and the second ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a first ID transmission channel; and the second ID transmission path of the 2i-1-th semiconductor chip and the first ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a second ID transmission channel.

[0011] In some embodiments, the logic chip is configured to: output a first ID signal to the ID decoding circuit of the 2i-1-th semiconductor chip through one of the first ID transmission channel and the second ID transmission channel; and output a second ID signal to the ID decoding circuit of the 2i-th semiconductor chip through the other of the first ID transmission channel and the second ID transmission channel, where the second ID signal is different from the first ID signal.

[0012] In some embodiments, the ID decoding circuit is configured to: decode the first ID signal to generate a first data selection signal, or decode the second ID signal to generate a second data

selection signal, where the second data selection signal is different from the first data selection signal.

[0013] In some embodiments, each of the semiconductor chips further includes: a first data receiving circuit, a second data receiving circuit, a first data transmission path, a second data transmission path, a first data selecting circuit, and a second data selecting circuit, where the first data transmission path and the second data transmission path penetrate through the substrate in a direction perpendicular to an active surface of one of the semiconductor chips and are symmetrically distributed based on the central axis; the first data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the first data receiving circuit; and the second data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the second data receiving circuit; and where the first data transmission path of the $2i-1$ -th semiconductor chip and the second data transmission path of the $2i$ -th semiconductor chip are sequentially connected to form a first data transmission channel; the second data transmission path of the $2i-1$ -th semiconductor chip and the first data transmission path of the $2i$ -th semiconductor chip are sequentially connected to form a second data transmission channel; the first data transmission channel is configured to transmit a first data signal; the second data transmission channel is configured to transmit a second data signal; and the second data signal is different from the first data signal.

[0014] In some embodiments, the first data selecting circuit of the $2i-1$ -th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the $2i-1$ -th semiconductor chip according to the first data selection signal; the second data selecting circuit of the $2i-1$ -th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the $2i-1$ -th semiconductor chip according to the first data selection signal; the first data selecting circuit of the $2i$ -th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the $2i$ -th semiconductor chip according to the second data selection signal; and the second data selecting circuit of the $2i$ -th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the $2i$ -th semiconductor chip according to the second data selection signal.

[0015] In some embodiments, central axes of any two adjacent semiconductor chips among the 1st to $2N$ -th semiconductor chips are aligned, where the first ID transmission path of one of the two adjacent semiconductor chips is aligned with the second ID transmission path of the other of the two adjacent semiconductor chips.

[0016] In some embodiments, an inactive face of a $2j$ -th semiconductor chip is bonded to an inactive face of a $2j+1$ -th semiconductor chip, where j is an integer greater than or equal to one, and $2j+1$ is less than $2N$.

[0017] In some embodiments, the first ID transmission path includes: a first connection pad located on an inactive face of the semiconductor chip, a first path located in the substrate, and a first interconnection structure located on the active surface of the semiconductor chip, where the first connection pad, the first path, and the first interconnection structure are sequentially connected; and the second ID transmission path includes: a second connection pad located on an inactive face of one of the semiconductor chips, a second path located in the substrate, and a second interconnection structure located on an active surface of the corresponding one of the semiconductor chips, where the second connection pad, the second path, and the second interconnection structure are sequentially connected.

[0018] According to a third aspect of the embodiments of the present disclosure, provided is a semiconductor device. The semiconductor device includes: a substrate, where the semiconductor chip according to any embodiment of the first aspect of the embodiments of the present disclosure is located on the substrate, and the semiconductor chip is bonded to the substrate; or, the

semiconductor structure according to any embodiment of the second aspect of the embodiments of the present disclosure is located on the substrate, and the semiconductor structure is bonded to the substrate.

[0019] In some embodiments, the semiconductor device includes: a high bandwidth memory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a schematic view of a three-dimensional stack structure according to an exemplary embodiment;

[0021] FIG. 2 is a cross-sectional view of a semiconductor chip according to an embodiment of the present disclosure;

[0022] FIG. 3 is a top view of a semiconductor chip according to an embodiment of the present disclosure;

[0023] FIG. 4 is a schematic view of a semiconductor chip according to an embodiment of the present disclosure;

[0024] FIG. 5 is a cross-sectional view of a semiconductor structure according to an embodiment of the present disclosure;

[0025] FIG. 6A is a first top view of a semiconductor chip according to an embodiment of the present disclosure; and

[0026] FIG. 6B is a second top view of a semiconductor chip according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0027] The technical solutions of the present disclosure will be further elaborated below with reference to the drawings and embodiments. While exemplary implementations of the present disclosure are shown in the drawings, it should be understood that the present disclosure may be implemented in various forms and should not be limited by the embodiments set forth herein.

Rather, these embodiments are provided so that the present disclosure will be more thoroughly understood and the scope of the present disclosure will be fully conveyed to those skilled in the art.

[0028] The present disclosure is more specifically described in the following paragraphs with reference to the drawings by way of example. The advantages and features of the present disclosure will become apparent from the following description and claims. It should be noted that the drawings are all in a very simplified form and not to a precise scale, and are provided only for the purpose of facilitating a convenient and clear description of the embodiments of the present disclosure.

[0029] It will be understood that the meaning of “on”, “above”, and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only includes the meaning of “on” something with no intermediate feature or layer therebetween (i.e., directly on something) but also includes the meaning of “on” something with an intermediate feature or a layer therebetween.

[0030] In the embodiments of the present disclosure, the terms “first”, “second”, “third”, and the like are used for distinguishing similar objects and are not necessarily used for describing a particular order or sequence.

[0031] In the embodiments of the present disclosure, the term “layer” refers to a material portion that includes a region having a thickness. A layer may extend over the entirety of the underlying or overlying structure or may have an extent that is less than the extent of the underlying or overlying structure. Furthermore, a layer may be a region of a homogeneous or inhomogeneous continuous structure having a thickness less than the thickness of a continuous structure. For example, a layer may be located between the top surface and the bottom surface of a continuous structure, or a layer may be located between any pair of horizontal planes at the top surface and the bottom surface of

the continuous structure. A layer may extend horizontally, perpendicularly, and/or along inclined surfaces. A layer may include a plurality of sub-layers.

[0032] It should be noted that unless conflicting, the technical solutions described in the embodiments of the present disclosure may be arbitrarily combined.

[0033] FIG. 1 is a schematic view of a three-dimensional stack structure according to an exemplary embodiment. Referring to FIG. 1, the stack structure includes a plurality of chips, e.g., chip 0, chip 1, chip 2, and chip 3, sequentially stacked on a logic chip. Here, four chips are merely examples to convey the present disclosure to those skilled in the art, and the number of the chip in the stack structure is not limited to four, but may be 8, 16, or more, which is not particularly limited in the present disclosure.

[0034] By assigning stack IDs to a plurality of chips, respectively, as shown in FIG. 1, an odd/even flag bit can be generated in each chip. The odd/even flag bit serves as a circuit direction switch to accurately transmit data to each chip. However, this method of assigning stack IDs is complicated.

[0035] In view of this, the embodiment of the present disclosure provides a semiconductor chip, a semiconductor structure, and a semiconductor device.

[0036] FIG. 2 is a schematic cross-sectional view of a semiconductor chip **100** according to an embodiment of the present disclosure, FIG. 3 is a top view of the semiconductor chip **100** according to an embodiment of the present disclosure, and FIG. 4 is a schematic view of the semiconductor chip **100** according to an embodiment of the present disclosure. The semiconductor chip **100** provided according to an embodiment of the present disclosure will be described in detail with reference to FIGS. 2, 3, and 4.

[0037] Referring to FIGS. 2 to 4, the semiconductor chip **100** includes a first ID transmission path **111**, a second ID transmission path **112**, and an ID decoding circuit **120**. The first ID transmission path **111** and the second ID transmission path **112** penetrate through a substrate in a direction perpendicular to the active surface **S1** of the semiconductor chip **100** and are symmetrically distributed based on the central axis **L** of the semiconductor chip **100**. The ID decoding circuit **120** is coupled to the first ID transmission path **111** or the second ID transmission path **112**, and the ID decoding circuit **120** is configured to receive an ID signal and decode the ID signal to generate a data selection signal. The semiconductor chip **100** outputs data signals transmitted in a plurality of data transmission paths to the corresponding internal receiving circuits respectively according to the data selection signal. It should be noted that the central axis is an axis passing through the center of the semiconductor chip and parallel to the active surface **S1** (or an inactive face **S2**), and is represented by a dotted line **L** in the present disclosure, as shown in FIG. 2 or 3.

[0038] The semiconductor chip **100** includes a front surface provided with a peripheral circuit and a memory cell array and a back surface opposite to the front surface. The front surface of the semiconductor chip **100** is denoted as the active surface **S1**, and the back surface of the semiconductor chip **100** is denoted as the inactive face **S2**, as shown in FIG. 2. The semiconductor chip **100** may be a dynamic random access memory (dynamic random access memory, DRAM) chip, or may be at least one of a static random access memory (static random access memory, SRAM) chip, a NOR flash memory chip, and a NAND flash memory chip. The embodiment of the present disclosure is not particularly limited to the type of the semiconductor chip **100**.

[0039] The first ID transmission path **111** and the second ID transmission path **112** are used to transmit ID signals. For example, the first ID transmission path **111** is configured to transmit a first ID signal **A1**, the second ID transmission path **112** is configured to transmit a second ID signal **A2**, and the second ID signal **A2** is different from the first ID signal **A1**.

[0040] It will be understood that when a plurality of semiconductor chips **100** are three-dimensionally stacked and packaged in a direction perpendicular to the horizontal plane, the ID signals transmitted in the first ID transmission path **111** and the second ID transmission path **112** may generate an odd flag bit/even flag bit in each semiconductor chip **100**. The odd flag bit or the even flag bit may serve as a switching ID of a transmission path between data and a circuit, thereby

ensuring that the data is accurately transmitted to the corresponding internal receiving circuit in each chip.

[0041] In a specific embodiment, the ID signal may be represented by a binary signal “0” or “1”. For example, the ID signal “0” is configured to indicate that the semiconductor chip **100** is an even chip in the stack structure, and the ID signal “1” is configured to indicate that the semiconductor chip **100** is an odd chip in the stack structure. For another example, the ID signal “0” is configured to indicate that the semiconductor chip **100** is an odd chip in the stack structure, and the ID signal “1” is configured to indicate that the semiconductor chip **100** is an even chip in the stack structure.

[0042] The ID decoding circuit is used for decoding the ID signal and generating a data selection signal. For example, the ID decoding circuit **120** is coupled to the first ID transmission path **111**, and the ID decoding circuit **120** is configured to receive the first ID signal A1 transmitted in the first ID transmission path **111** and decode the first ID signal A1 to generate the data selection signal. As another example, the ID decoding circuit **120** is coupled to the second ID transmission path **112**, and the ID decoding circuit **120** is configured to receive the second ID signal A2 transmitted in the second ID transmission path **112** and decode the second ID signal A2 to generate the data selection signal.

[0043] It should be noted that since the second ID signal A2 is different from the first ID signal A1, the data selection signal generated by decoding the first ID signal is different from the data selection signal generated by decoding the second ID signal in the embodiment. In other embodiments, when the second ID signal is the same as the first ID signal, the data selection signal generated by decoding the first ID signal may be the same as the data selection signal generated by decoding the second ID signal.

[0044] The semiconductor chip **100** includes a plurality of data transmission paths and a plurality of internal receiving circuits. Each internal receiving circuit is separately coupled to the plurality of data transmission paths, and each internal receiving circuit may receive the data signal transmitted in one of the plurality of data transmission paths according to the data selection signal. Here, the number of the data transmission path may be two or more, and the number of the internal receiving circuit may be two or more, which is not particularly limited in the present disclosure.

[0045] In the embodiments of the present disclosure, by coupling the ID decoding circuit to the first ID transmission path or the second ID transmission path, the ID decoding circuit may generate the data selection signal according to the ID signal transmitted in the first ID transmission path or the second ID transmission path, and the internal receiving circuit may receive the data signal transmitted in one of the plurality of data transmission paths according to the data selection signal, thereby ensuring that the data is accurately transmitted to the corresponding internal receiving circuit in the semiconductor chip.

[0046] In addition, the first ID transmission path and the second ID transmission path are symmetrically distributed based on the central axis of the semiconductor chip; when a plurality of semiconductor chips are stacked to form a three-dimensional stack structure, the ID signals transmitted in the first ID transmission path and the second ID transmission path may generate an odd flag bit/even flag bit in each semiconductor chip, such that the complexity of assigning the stack ID is reduced, and meanwhile, it can also ensure that the data is separately and accurately transmitted to each chip of the three-dimensional stack structure through the same transmission paths.

[0047] In some embodiments, with reference to FIGS. 2 and 4, the internal receiving circuit includes: a first data receiving circuit **131** and a second data receiving circuit **132**. The first data receiving circuit **131** is coupled to a first data selecting circuit **141**; the second data receiving circuit **132** is coupled to a second data selecting circuit **142**.

[0048] The plurality of data transmission paths include: [0049] a first data transmission path **151**, penetrating through the substrate in the direction perpendicular to the active surface S1 of the semiconductor chip **100**, where the first data transmission path **151** is configured to transmit a first

data signal **B1**; [0050] a second data transmission path **152**, penetrating through the substrate in the direction perpendicular to the active surface **S1** of the semiconductor chip **100**, and symmetrically distributed with the first data transmission path **151** based on the central axis **L**, where the second data transmission path **152** is configured to transmit a second data signal **B2**; the second data signal **B2** is different from the first data signal **B1**; [0051] a first data selecting circuit **141**, separately coupled to the ID decoding circuit **120**, the first data transmission path **151**, and the second data transmission path **152**, where the first data selecting circuit **141** is configured to output one of the first data signal **B1** or the second data signal **B2** to the first data receiving circuit **131** according to the data selection signal; and [0052] a second data selecting circuit **142**, separately coupled to the ID decoding circuit **120**, the first data transmission path **151**, and the second data transmission path **152**, where the second data selecting circuit **142** is configured to output the other of the first data signal **B1** or the second data signal **B2** to the second data receiving circuit **132** according to the data selection signal.

[0053] The first data selecting circuit **141** and the second data selecting circuit **142** may output corresponding data signals according to the data selection signals. Taking the first data selecting circuit **141** as an example, the first data selecting circuit **141** includes a control terminal, a plurality of input terminals, and an output terminal. The control terminal of the first data selecting circuit **141** is configured to receive the data selection signal, the plurality of input terminals of the first data selecting circuit **141** are respectively coupled to the plurality of data transmission paths, and the output terminal of the first data selecting circuit **141** is coupled to the first data receiving circuit **131**; the first data selecting circuit **141** may output the first data signal **B1** (or the second data signal **B2**) to the first data receiving circuit **131** under the control of the data selection signal received by the control terminal.

[0054] Similarly, the second data selecting circuit **142** includes a control terminal, a plurality of input terminals, and an output terminal. The control terminal of the second data selecting circuit **142** is configured to receive the data selection signal, the plurality of input terminals of the second data selecting circuit **142** are respectively coupled to the plurality of data transmission paths, and the output terminal of the second data selecting circuit **142** is coupled to the second data receiving circuit **132**; the second data selecting circuit **142** may output the second data signal **B2** (or the first data signal **B1**) to the second data receiving circuit **132** under the control of the data selection signal received by the control terminal.

[0055] In some embodiments, the first data signal **B1** and the second data signal **B2** may be different row address (row address, RA) signals, for example, a row address signal **RA1** and a row address signal **RA2**. The binary values corresponding to the row address signal **RA1** and the row address signal **RA2** may be the same or different, which is not particularly limited in the present disclosure. In other embodiments, the first data signal **B1** and the second data signal **B2** may also be signals of different types. The first data signal **B1** and the second data signal **B2** are respectively transmitted to different data receiving circuits in the same semiconductor chip.

[0056] It should be noted that in the embodiment, the first data selecting circuit **141** and the second data selecting circuit **142** are different data selecting circuits, and the first data selecting circuit **141** and the second data selecting circuit **142** may output data signals of different types based on the same data selection signal. For example, the first data selecting circuit **141** outputs the row address signal **RA1** to the first data receiving circuit **131**, and the second data selecting circuit **142** outputs the row address signal **RA2** to the second data receiving circuit **132**.

[0057] The first data transmission path **151** and the second data transmission path **152** may be a through silicon via (through silicon via, TSV) structure in the semiconductor chip **100**.

[0058] In the embodiment of the present disclosure, through the arrangement of the first data selecting circuit and the second data selecting circuit, the first data selecting circuit may transmit the first data signal (or the second data signal) to the first data receiving circuit according to the data selection signal, and the second data selecting circuit may transmit the second data signal (or

the first data signal) to the second data receiving circuit according to the data selection signal, thereby ensuring that the data is accurately transmitted to each data receiving circuit in the semiconductor chip.

[0059] In some embodiments, referring to FIG. 2, the first ID transmission path **111** includes: a first connection pad **161** located on the inactive face **S2** of the semiconductor chip **100**, a first path **171** located in the substrate, and a first interconnection structure **181** located on the active surface **S1** of the semiconductor chip **100**, where the first connection pad **161**, the first path **171**, and the first interconnection structure **181** are sequentially connected; the second ID transmission path **112** includes: a second connection pad **162** located on the inactive face **S2** of the semiconductor chip **100**, a second path **172** located in the substrate, and a second interconnection structure **182** located on the active surface **S1** of the semiconductor chip **100**, where the second connection pad **162**, the second path **172**, and the second interconnection structure **182** are sequentially connected.

[0060] In a specific embodiment, the first connection pad **161** and the second connection pad **162** may be micro bumps (uBumps) located on the inactive face **S2** of the semiconductor chip **100**; the first path **171** and the second path **172** may be TSV structures; the first interconnection structure **181** and the second interconnection structure **182** may be metal wirings located on the active surface **S1** of the semiconductor chip **100**.

[0061] It should be noted that the number of the micro bump, the number of the through silicon via structure, and the number of the metal wiring shown in FIG. 2 are exemplary to convey the present disclosure to those skilled in the art. In practical applications, the number of the micro bump, the number of the through silicon via structure, and the number of the metal wiring in the semiconductor chip are not limited thereto.

[0062] In some embodiments, the first connection pad **161** and the second connection pad **162** are symmetrically distributed based on the central axis **L**; the first path **171** and the second path **172** are symmetrically distributed based on the central axis **L**; the first interconnection structure **181** and the second interconnection structure **182** are symmetrically distributed based on the central axis **L**.

[0063] In some embodiments, the ID decoding circuit **120** is coupled to the first interconnection structure **181** or the second interconnection structure **182**.

[0064] The materials of the first connection pad **161**, the second connection pad **162**, the first path **171**, the second path **172**, the first interconnection structure **181**, and the second interconnection structure **182** include a conductive material, for example, any one of polycrystalline silicon, doped polycrystalline silicon, tantalum, titanium, tungsten, copper, titanium nitride, tantalum nitride, or tungsten nitride, or a combination thereof. The materials of any two of the first connection pad **161**, the second connection pad **162**, the first path **171**, the second path **172**, the first interconnection structure **181**, and the second interconnection structure **182** may be the same or different.

[0065] In some embodiments, the first data transmission path **151** includes: a third connection pad located on the inactive face of the semiconductor chip, a third path located in the substrate, and a third interconnection structure located on the active surface of the semiconductor chip, where the third connection pad, the third path, and the third interconnection structure are sequentially connected; the second data transmission path **152** includes: a fourth connection pad located on the inactive face of the semiconductor chip, a fourth path located in the substrate, and a fourth interconnection structure located on the active surface of the semiconductor chip, where the fourth connection pad, the fourth path, and the fourth interconnection structure are sequentially connected. Here, the structures of the first data transmission path **151** and the second data transmission path **152** may refer to the first ID transmission path **111** and the second ID transmission path **112**, which will not be repeated here.

[0066] In some embodiments, referring to FIG. 3, the semiconductor chip **100** further includes a first redundant path **191** and a second redundant path **192**, and the first redundant path **191** and the second redundant path **192** are symmetrically distributed based on the central axis **L** of the semiconductor chip. The number of the first redundant path **191** and the second redundant path **192**

may be one or more, and three first redundant paths **191** and three second redundant paths **192** are shown in FIG. **3**.

[0067] Based on the above semiconductor chip, the embodiment of the present disclosure further provides a semiconductor structure.

[0068] FIG. **5** is a schematic view of a semiconductor structure **300** according to an embodiment of the present disclosure, and FIGS. **6A** and **6B** are top views of a semiconductor chip according to an embodiment of the present disclosure. The semiconductor structure **300** provided according to an embodiment of the present disclosure will be described in detail with reference to FIGS. **5**, **6A**, and **6B**.

[0069] Referring to FIG. **5**, the semiconductor structure **300** includes: a logic chip **200** and a stack structure stacked on the logic chip. The stack structure includes: the 1st to 2N-th semiconductor chips stacked sequentially, where the 1st to 2N-th semiconductor chips are the same memory chips; the inactive face of the 1st semiconductor chip is bonded to the logic chip, and the active surface of the 2i-th semiconductor chip is bonded to the active surface of the 2i-1-th semiconductor chip; N is a positive integer, and i is a positive integer less than or equal to N.

[0070] The semiconductor chip includes: a first ID transmission path, a second ID transmission path, and an ID decoding circuit. The first ID transmission path and the second ID transmission path penetrate through the substrate of the semiconductor chip in a direction perpendicular to the active surface of the semiconductor chip and are symmetrically distributed based on the central axis of the semiconductor chip; the ID decoding circuit is coupled to the first ID transmission path or the second ID transmission path, and the ID decoding circuit is configured to receive an ID signal, which indicates that the semiconductor chip is the 2i-th semiconductor chip or the 2i-1-th semiconductor chip in the stack structure.

[0071] Here, the first ID transmission path of the 2i-1-th semiconductor chip and the second ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a first ID transmission channel **301**; the second ID transmission path of the 2i-1-th semiconductor chip and the first ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a second ID transmission channel **302**.

[0072] The semiconductor structure **300** may be a memory including a logic chip and a memory chip, e.g., a high bandwidth memory (high bandwidth memory, HBM), and the semiconductor structure **300** may also be other memories known in the art.

[0073] The logic chip **200** may be one or more processors that communicate with the memory chip to access data from the memory chip or to store data in the memory chip. The logic chip **200** includes: a graphics processing unit (GPU), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a central processing unit (CPU), or other known electronic circuits serving as processors.

[0074] The stack structure includes a plurality of semiconductor chips stacked sequentially. For example, FIG. **5** shows four semiconductor chips, which are denoted as the 1st semiconductor chip **100a**, the 2nd semiconductor chip **100b**, the 3rd semiconductor chip **100c**, and the 4th semiconductor chip **100d**, respectively. The 1st semiconductor chip **100a**, the 2nd semiconductor chip **100b**, the 3rd semiconductor chip **100c**, and the 4th semiconductor chip **100d** are the same memory chips. Here, the 1st to 4th semiconductor chips may be explained with reference to the semiconductor chip **100** shown in FIG. **2**. It should be noted that the number of the semiconductor chip in the stack structure is not limited to four shown in FIG. **5**, and may also be 8, 16, or more.

[0075] The 1st semiconductor chip **100a** includes a first ID transmission path **111a**, a second ID transmission path **112a**, and an ID decoding circuit **120a**; the 2nd semiconductor chip **100b** includes a first ID transmission path **111b**, a second ID transmission path **112b**, and an ID decoding circuit **120b**; the 3rd semiconductor chip **100c** includes a first ID transmission path **111c**, a second ID transmission path **112c**, and an ID decoding circuit **120c**; the 4th semiconductor chip **100d** includes a first ID transmission path **111d**, a second ID transmission path **112d**, and an ID decoding

circuit **120d**. Here, the first ID transmission paths **111a** to **111d**, the second ID transmission paths **112a** to **112d**, and the ID decoding circuits **120a** to **120d** may be separately explained with reference to the first ID transmission path **111**, the second ID transmission path **112**, and the ID decoding circuit **120** shown in FIG. 2, which will not be repeated here.

[0076] Referring to FIG. 5, the 1st semiconductor chip set **300a** may be formed by bonding the active surface of the 1st semiconductor chip **100a** to the active surface of the 2nd semiconductor chip **100b**; the 2nd semiconductor chip set **300b** may be formed by bonding the active surface of the 3rd semiconductor chip **100c** to the active surface of the 4th semiconductor chip **100d**; 2N semiconductor chips may form N semiconductor chip sets.

[0077] It should be noted that the number of the semiconductor chip set in the semiconductor structure may be one or more. Two semiconductor chips in a semiconductor chip set are face-to-face bonded (the active surface is bonded to the active surface); for example: the 1st semiconductor chip **100a** and the 2nd semiconductor chip **100b**, and the 3rd semiconductor chip **100c** and the 4th semiconductor chip **100d** may be bonded by hybrid bonding, such as fusion bonding, diffusion bonding, or eutectic bonding. Two adjacent semiconductor chip sets are back-to-back bonded (the inactive face is bonded to the inactive face); for example: the 2nd semiconductor chip **100b** in the 1st semiconductor chip set **300a** and the 3rd semiconductor chip **100c** in the 2nd semiconductor chip set **300b** may be connected by a micro bump (uBump). In other embodiments, two semiconductor chips in a semiconductor chip set may also be bonded by other bonding methods known in the art. Two semiconductor chips in the 1st semiconductor chip set will be described below with reference to FIGS. 6A and 6B.

[0078] FIG. 6A shows top views of the 1st semiconductor chip **100a** and the 2nd semiconductor chip **100b** separately. Referring to FIG. 6A, the 1st semiconductor chip **100a** includes an upper transmission region located on one side of the central axis L and a lower transmission region located on the other side of the central axis L, and the 2nd semiconductor chip **100b** includes an upper transmission region located on one side of the central axis L and a lower transmission region located on the other side of the central axis L. Here, the semiconductor chip may be divided into two equal-sized parts based on the central axis L; the part where the first ID transmission path **111** is located may be defined as the upper transmission region, and the part where the second ID transmission path **112** is located may be defined as the lower transmission region.

[0079] In other embodiments, the semiconductor chip may be divided into two equal-sized parts based on the central axis P; the part where the first ID transmission path **111** is located may be defined as a left transmission region, and the part where the second ID transmission path **112** is located may be defined as a right transmission region, as shown in FIG. 6B. Those skilled in the art can make a selection according to actual needs. Here, the central axis P passes through the center of the semiconductor chip and is parallel to the active surface, and the central axis P is perpendicular to and intersects with the central axis L.

[0080] After the active surface of the 1st semiconductor chip **100a** is bonded to the active surface of the 2nd semiconductor chip **100b**, the lower transmission region of the 2nd semiconductor chip **100b** is located above the upper transmission region of the 1st semiconductor chip **100a**, and the upper transmission region of the 2nd semiconductor chip **100b** is located above the lower transmission region of the 1st semiconductor chip **100a**, as shown in FIG. 5. That is, after the active surface of the 1st semiconductor chip **100a** is bonded to the active surface of the 2nd semiconductor chip **100b**, the ID decoding circuit **120a** of the 1st semiconductor chip **100a** and the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** are respectively located on both sides of the symmetry plane M of the semiconductor structure **300**, and the same data receiving circuit (e.g., **131a** and **131b**) in each semiconductor chip is separately located on both sides of the symmetry plane M, where the symmetry plane M is perpendicular to the active surface S1 (or the inactive face S2), and the central axis L is located in the symmetry plane M.

[0081] When the semiconductor structure **300** includes the 1st semiconductor chip set, the first ID

transmission channel **301** includes the first ID transmission path **111a** of the 1st semiconductor chip **100a** and the second ID transmission path **112b** of the 2nd semiconductor chip **100b**; the second ID transmission channel **302** includes the second ID transmission path **112a** of the 1st semiconductor chip **100a** and the first ID transmission path **111b** of the 2nd semiconductor chip **100b**. Further, when the semiconductor structure **300** further includes a 2nd semiconductor chip set, the first ID transmission channel **301** further includes the first ID transmission path **111c** of the 3rd semiconductor chip **100c** and the second ID transmission path **112d** of the 4th semiconductor chip **100d**; the second ID transmission channel **302** further includes the second ID transmission path **112c** of the 3rd semiconductor chip **100c** and the first ID transmission path **111d** of the 4th semiconductor chip **100d**.

[0082] Here, the sequentially connected ID transmission paths located on one side of the symmetry plane M form the first ID transmission channel **301**, and the sequentially connected ID transmission paths located on the other side of the symmetry plane M form the second ID transmission channel **302**; the first ID transmission channel **301** and the second ID transmission channel **302** are symmetrically distributed based on the central axis L.

[0083] The first ID transmission channel **301** and the second ID transmission channel **302** are used for transmitting ID signals. For example, the first ID transmission channel **301** is configured to transmit the first ID signal, and the second ID transmission channel **302** is configured to transmit the second ID signal; the first ID signal is different from the second ID signal. The first ID signal may be represented by a binary signal “0” and the second ID signal may be represented by a binary signal “1”.

[0084] In some embodiments, when the ID decoding circuit is coupled to the first ID transmission path, the ID decoding circuit **120a** of the 1st semiconductor chip **100a** may assign an odd flag bit to the 1st semiconductor chip **100a** according to the first ID signal “0” transmitted in the first ID transmission channel **301**, and the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** may assign an even flag bit to the 2nd semiconductor chip **100b** according to the second ID signal “1” transmitted in the second ID transmission channel **302**. Further, the ID decoding circuit **120c** of the 3rd semiconductor chip **100c** may assign an odd flag bit to the 3rd semiconductor chip **100c** according to the first ID signal “0” transmitted in the first ID transmission channel **301**, and the ID decoding circuit **120d** of the 4th semiconductor chip **100d** may assign an even flag bit to the 4th semiconductor chip **100d** according to the second ID signal “1” transmitted in the second ID transmission channel **302**.

[0085] In some other embodiments, when the ID decoding circuit is coupled to the second ID transmission path, the ID decoding circuit **120a** of the 1st semiconductor chip **100a** may assign an odd flag bit to the 1st semiconductor chip **100a** according to the second ID signal “1” transmitted in the second ID transmission channel **302**, and the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** may assign an even flag bit to the 2nd semiconductor chip **100b** according to the first ID signal “0” transmitted in the first ID transmission channel **301**. Further, the ID decoding circuit **120c** of the 3rd semiconductor chip **100c** may assign an odd flag bit to the 3rd semiconductor chip **100c** according to the second ID signal “1” transmitted in the second ID transmission channel **302**, and the ID decoding circuit **120d** of the 4th semiconductor chip **100d** may assign an even flag bit to the 4th semiconductor chip **100d** according to the first ID signal “0” transmitted in the first ID transmission channel **301**.

[0086] It will be understood that in the above two embodiments, different definitions may be assigned to the ID signal according to the position of the ID decoding circuit. For example, the ID signal “0” indicates that the semiconductor chip is a memory chip in an odd tier of the stack structure, and the ID signal “1” indicates that the semiconductor chip is a memory chip in an even tier of the stack structure; alternatively, the ID signal “0” indicates that the semiconductor chip is a memory chip in an even tier of the stack structure, and the ID signal “1” indicates that the semiconductor chip is a memory chip in an odd tier of the stack structure.

[0087] In other embodiments, fixed definitions may also be assigned to the ID signals “0” and “1”, and the ID signals transmitted in the first ID transmission channel **301** and the second ID transmission channel **302** are exchanged according to the position of the ID decoding circuit, thereby ensuring that each semiconductor chip in the stack structure is assigned an odd flag bit or an even flag bit through the first ID transmission channel **301** and the second ID transmission channel **302**.

[0088] It should be noted that since the same data receiving circuit (e.g., **131a** and **131b**) in the semiconductor chip **100a** located in an odd tier and the semiconductor chip **100b** located in an even tier are respectively located on both sides of the symmetry plane M, and the data signal transmitted in the data transmission channel is not switched, an odd-even ID signal is required to switch the data signal transmitted to the same data receiving circuit. According to the embodiment of the present disclosure, each semiconductor chip in the stack structure is assigned an odd flag bit or an even flag bit, and the odd flag bit or the even flag bit may serve as a switching ID of a transmission path between data and a circuit, such that the data is accurately transmitted to the corresponding data receiving circuit in each chip of the stack structure.

[0089] In the embodiment of the present disclosure, a stack structure is arranged on a logic chip; the stack structure includes the 1st to 2N-th semiconductor chips stacked sequentially, where the 1st to 2N-th semiconductor chips are the same memory chips; the first ID transmission path of the 2i-1-th semiconductor chip and the second ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a first ID transmission channel, and the second ID transmission path of the 2i-1-th semiconductor chip and the first ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a second ID transmission channel; through the first ID transmission channel and the second ID transmission channel, each semiconductor chip in the stack structure can be assigned an odd flag bit or an even flag bit, and the odd flag bit or the even flag bit may serve as a switching ID of a transmission path between data and a circuit, such that the data is accurately transmitted to the corresponding data receiving circuit in each chip of the stack structure.

[0090] In addition, according to the embodiment of the present disclosure, the first ID transmission path and the second ID transmission path are symmetrically distributed based on the central axis in the semiconductor chip, such that the complexity of assigning the stack ID is reduced, thereby allowing more memory chips to be three-dimensionally packaged, which is beneficial to improving the integration level of the semiconductor structure and increasing the memory capacity.

[0091] In some embodiments, referring to FIG. 5, the logic chip **200** is configured to: output the first ID signal to the ID decoding circuit of the 2i-1-th semiconductor chip through one of the first ID transmission channel **301** and the second ID transmission channel **302**; and output the second ID signal to the ID decoding circuit of the 2i-th semiconductor chip through the other of the first ID transmission channel and the second ID transmission channel, where the second ID signal is different from the first ID signal.

[0092] The logic chip **200** includes an ID signal generation circuit for generating an ID signal. For example, the ID signal generation circuit includes a first sub-ID signal generation circuit for generating the first ID signal (or the second ID signal) and a second sub-ID signal generation circuit for generating the second ID signal (or the first ID signal); the first sub-ID signal generation circuit may be coupled to the first ID transmission channel **301**, and the second sub-ID signal generation circuit may be coupled to the second ID transmission channel **302**.

[0093] In some embodiments, when the ID decoding circuit in each chip is coupled to the first ID transmission path, the logic chip **200** separately transmits the first ID signal “0” to the ID decoding circuit **120a** of the 1st semiconductor chip **100a** and the ID decoding circuit **120c** of the 3rd semiconductor chip **100c** through the first ID transmission channel **301**, and the logic chip **200** separately transmits the second ID signal “1” to the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** and the ID decoding circuit **120d** of the 4th semiconductor chip **100d**

through the second ID transmission channel **302**.

[0094] In some other embodiments, when the ID decoding circuit in each chip is coupled to the second ID transmission path, the logic chip **200** separately transmits the second ID signal “1” to the ID decoding circuit **120a** of the 1st semiconductor chip **100a** and the ID decoding circuit **120c** of the 3rd semiconductor chip **100c** through the second ID transmission channel **302**, and the logic chip **200** separately transmits the first ID signal “0” to the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** and the ID decoding circuit **120d** of the 4th semiconductor chip **100d** through the first ID transmission channel **301**.

[0095] It will be understood that according to the semiconductor structure provided in the embodiment of the present disclosure, different ID signals (for example, “0” and “1”) may be transmitted to the ID decoding circuits of the memory chips in different tiers, respectively, and the ID decoding circuits decode the received ID signals to generate the data selection signals, such that the data is accurately transmitted to each chip of the stack structure.

[0096] In some embodiments, the ID decoding circuit is configured to: decode the first ID signal to generate a first data selection signal, or decode the second ID signal to generate a second data selection signal, where the second data selection signal is different from the first data selection signal.

[0097] In an embodiment, the ID decoding circuit in each chip is coupled to the first ID transmission path. With reference to FIG. 5, the ID decoding circuit **120a** of the 1st semiconductor chip **100a** receives the first ID signal “0” transmitted by the first ID transmission path **111a** and decodes the first ID signal to generate the first data selection signal, for example, the first data selection signal “1”; the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** receives the second ID signal “1” transmitted by the first ID transmission path **111b** and decodes the second ID signal to generate the second data selection signal, for example, the second data selection signal “0”.

[0098] In another embodiment, the ID decoding circuit in each chip is coupled to the second ID transmission path. With reference to FIG. 5, the ID decoding circuit **120a** of the 1st semiconductor chip **100a** receives the second ID signal “1” transmitted by the second ID transmission path **112a** and decodes the second ID signal to generate the second data selection signal, for example, the second data selection signal “0”; the ID decoding circuit **120b** of the 2nd semiconductor chip **100b** receives the first ID signal “0” transmitted by the second ID transmission path **112b** and decodes the first ID signal to generate the first data selection signal, for example, the first data selection signal “1”.

[0099] In some embodiments, the semiconductor chip further includes: a first data receiving circuit, a second data receiving circuit, a first data transmission path, a second data transmission path, a first data selecting circuit, and a second data selecting circuit. The first data transmission path and the second data transmission path penetrate through the substrate in the direction perpendicular to the active surface of the semiconductor chip and are symmetrically distributed based on the central axis. The first data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the first data receiving circuit; the second data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the second data receiving circuit.

[0100] Here, the first data transmission path of the 2i-1-th semiconductor chip and the second data transmission path of the 2i-th semiconductor chip are sequentially connected to form a first data transmission channel **311**; the second data transmission path of the 2i-1-th semiconductor chip and the first data transmission path of the 2i-th semiconductor chip are sequentially connected to form a second data transmission channel **312**. The first data transmission channel **311** is configured to transmit a first data signal, the second data transmission channel **312** is configured to transmit a second data signal, and the second data signal is different from the first data signal.

[0101] Referring to FIG. 5, the 1st semiconductor chip **100a** includes a first data receiving circuit

131a, a second data receiving circuit **132a**, a first data transmission path **151a**, a second data transmission path **152a**, a first data selecting circuit **141a**, and a second data selecting circuit **142a**; the 2nd semiconductor chip **100b** includes a first data receiving circuit **131b**, a second data receiving circuit **132b**, a first data transmission path **151b**, a second data transmission path **152b**, a first data selecting circuit **141b**, and a second data selecting circuit **142b**; the 3rd semiconductor chip **100c** includes a first data receiving circuit **131c**, a second data receiving circuit **132c**, a first data transmission path **151c**, a second data transmission path **152c**, a first data selecting circuit **141c**, and a second data selecting circuit **142c**; the 4th semiconductor chip **100d** includes a first data receiving circuit **131d**, a second data receiving circuit **132d**, a first data transmission path **151d**, a second data transmission path **152d**, a first data selecting circuit **141d**, and a second data selecting circuit **142d**.

[0102] The first data receiving circuits **131a** to **131d**, the second data receiving circuits **132a** to **132d**, the first data transmission paths **151a** to **151d**, the second data transmission paths **152a** to **152d**, the first data selecting circuits **141a** to **141d**, and the second data selecting circuits **142a** to **142d** can be explained separately with reference to the first data receiving circuit **131**, the second data receiving circuit **132**, the first data transmission path **151**, the second data transmission path **152**, the first data selecting circuit **141**, and the second data selecting circuit **142** shown in FIGS. 2 and 4, which will not be repeated here.

[0103] The first data transmission channel **311** includes the first data transmission path **151a** of the 1st semiconductor chip **100a**, the second data transmission path **152b** of the 2nd semiconductor chip **100b**, the first data transmission path **151c** of the 3rd semiconductor chip **100c**, and the second data transmission path **152d** of the 4th semiconductor chip **100d**; the second data transmission channel **312** includes the second data transmission path **152a** of the 1st semiconductor chip **100a**, the first data transmission path **151b** of the 2nd semiconductor chip **100b**, the second data transmission path **152c** of the 3rd semiconductor chip **100c**, and the first data transmission path **151d** of the 4th semiconductor chip **100d**.

[0104] It will be understood that the sequentially connected data transmission paths located on one side of the symmetry plane M form the first data transmission channel **311**, and the sequentially connected data transmission paths located on the other side of the symmetry plane M form the second data transmission channel **312**; the first data transmission channel **311** and the second data transmission channel **312** are symmetrically distributed based on the central axis L.

[0105] The first data transmission channel **311** and the second data transmission channel **312** are used for transmitting data signals. For example, the first data transmission channel is configured to transmit a first data signal B1, and the second data transmission channel is configured to transmit a second data signal B2.

[0106] In some embodiments, the first data selecting circuit of the 2i-1-th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the 2i-1-th semiconductor chip according to the first data selection signal.

[0107] The second data selecting circuit of the 2i-1-th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the 2i-1-th semiconductor chip according to the first data selection signal.

[0108] The first data selecting circuit of the 2i-th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the 2i-th semiconductor chip according to the second data selection signal.

[0109] The second data selecting circuit of the 2i-th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the 2i-th semiconductor chip according to the second data selection signal.

[0110] The first data selecting circuit **141a** of the 1st semiconductor chip **100a** may output the first data signal B1 to the first data receiving circuit **131a** of the 1st semiconductor chip **100a** according to the first data selection signal; the second data selecting circuit **142a** of the 1st semiconductor

chip **100a** may output the second data signal B2 to the second data receiving circuit **132a** of the 1st semiconductor chip **100a** according to the first data selection signal.

[0111] The first data selecting circuit **141b** of the 2nd semiconductor chip **100b** may output the first data signal B1 to the first data receiving circuit **131b** of the 2nd semiconductor chip **100b** according to the second data selection signal; the second data selecting circuit **142b** of the 2nd semiconductor chip **100b** may output the second data signal B2 to the second data receiving circuit **132b** of the 2nd semiconductor chip **100b** according to the second data selection signal.

[0112] The first data selecting circuit **141c** of the 3rd semiconductor chip **100c** may output the first data signal B1 to the first data receiving circuit **131c** of the 3rd semiconductor chip **100c** according to the first data selection signal; the second data selecting circuit **142c** of the 3rd semiconductor chip **100c** may output the second data signal B2 to the second data receiving circuit **132c** of the 3rd semiconductor chip **100c** according to the first data selection signal.

[0113] The first data selecting circuit **141d** of the 4th semiconductor chip **100d** may output the first data signal B1 to the first data receiving circuit **131d** of the 4th semiconductor chip **100d** according to the second data selection signal; the second data selecting circuit **142d** of the 4th semiconductor chip **100d** may output the second data signal B2 to the second data receiving circuit **132d** of the 4th semiconductor chip **100d** according to the second data selection signal.

[0114] It should be noted that when the 1st semiconductor chip **100a**, the 2nd semiconductor chip **100b**, the 3rd semiconductor chip **100c**, and the 4th semiconductor chip **100d** are not separately assigned an odd flag bit and an even flag bit, the first data receiving circuit of the semiconductor chip located in an odd tier receives the first data signal B1, and the second data receiving circuit of the semiconductor chip located in an odd tier receives the second data signal B2; however, the first data receiving circuit of the semiconductor chip located in an even tier receives the second data signal B2, and the second data receiving circuit of the semiconductor chip located in an even tier receives the first data signal B1, such that the first data signal B1 and the second data signal B2 cannot be accurately transmitted to the corresponding internal receiving circuits.

[0115] According to the semiconductor structure provided in the embodiment of the present disclosure, each semiconductor chip in the stack structure is separately assigned an odd flag bit or an even flag bit, and the odd flag bit or the even flag bit may serve as a switching ID of a transmission path between data and a circuit, such that the data is accurately transmitted to the corresponding internal receiving circuit in each semiconductor chip of the stack structure.

[0116] In some embodiments, the central axes of any two adjacent semiconductor chips among the 1st to 2N-th semiconductor chips are aligned, where the first ID transmission path of one of the two adjacent semiconductor chips is aligned with the second ID transmission path of the other of the two adjacent semiconductor chips.

[0117] Referring to FIG. 5, the central axes L of the 1st semiconductor chip **100a**, the 2nd semiconductor chip **100b**, the 3rd semiconductor chip **100c**, and the 4th semiconductor chip **100d** are all aligned and coincide.

[0118] The first ID transmission path **111a** of the 1st semiconductor chip **100a** and the second ID transmission path **112b** of the 2nd semiconductor chip **100b** are aligned, and the second ID transmission path **112a** of the 1st semiconductor chip **100a** and the first ID transmission path **111b** of the 2nd semiconductor chip **100b** are aligned. In a specific embodiment, the first interconnection structure of the 1st semiconductor chip **100a** and the second interconnection structure of the 2nd semiconductor chip **100b** are aligned, and the second interconnection structure of the 1st semiconductor chip **100a** and the first interconnection structure of the 2nd semiconductor chip **100b** are aligned.

[0119] The second ID transmission path **112b** of the 2nd semiconductor chip **100b** and the first ID transmission path **111c** of the 3rd semiconductor chip **100c** are aligned, and the first ID transmission path **111b** of the 2nd semiconductor chip **100b** and the second ID transmission path **112c** of the 3rd semiconductor chip **100c** are aligned. In a specific embodiment, the second

connection pad of the 2nd semiconductor chip **100b** and the first connection pad of the 3rd semiconductor chip **100c** are aligned, and the first connection pad of the 2nd semiconductor chip **100b** and the second connection pad of the 3rd semiconductor chip **100c** are aligned.

[0120] In some embodiments, the inactive face of the 2j-th semiconductor chip is bonded to the inactive face of the 2j+1-th semiconductor chip, where j is an integer greater than or equal to one, and 2j+1 is less than 2N. For example, as shown in FIG. 5, the inactive face of the 2nd semiconductor chip **100b** is bonded to the inactive face of the 3rd semiconductor chip **100c**, e.g., being bonded through the connection pad on the inactive face of the semiconductor chip.

[0121] In some embodiments, the first ID transmission path includes: a first connection pad located on the inactive face of the semiconductor chip, a first path located in the substrate, and a first interconnection structure located on the active surface of the semiconductor chip, where the first connection pad, the first path, and the first interconnection structure are sequentially connected. The second ID transmission path includes: a second connection pad located on the inactive face of the semiconductor chip, a second path located in the substrate, and a second interconnection structure located on the active surface of the semiconductor chip, where the second connection pad, the second path, and the second interconnection structure are sequentially connected. The first ID transmission path and the second ID transmission path may refer to the first ID transmission path **111** and the second ID transmission path **112** in FIG. 2, respectively, which will not be repeated here.

[0122] Based on the above semiconductor chip or semiconductor structure, an embodiment of the present disclosure further provides a semiconductor device, including: [0123] a substrate.

[0124] The semiconductor chip **100** according to any one of the above embodiments is located on the substrate, and the semiconductor chip **100** is bonded to the substrate. Alternatively, the semiconductor structure **300** according to any one of the above embodiments is located on the substrate, and the semiconductor structure **300** is bonded to the substrate.

[0125] The substrate includes a package substrate for bearing a logic chip and a memory chip, for example, a low temperature co-fired ceramic substrate or a printed circuit board.

[0126] In some embodiments, the semiconductor device includes: a high bandwidth memory.

[0127] The above description is only the specific embodiments of the present disclosure, but the protection scope of the present disclosure is not limited thereto; changes or substitutions that any of those skilled in the art can easily think of within the technical scope disclosed by the present disclosure shall all fall within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be defined by the protection scope of the claims.

Claims

1. A semiconductor chip, comprising: a first identification (ID) transmission path, penetrating through a substrate of the semiconductor chip in a direction perpendicular to an active surface of the semiconductor chip; a second ID transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, and symmetrically distributed with the first ID transmission path based on a central axis of the semiconductor chip, wherein the central axis passes through a center of the semiconductor chip and is parallel to the active surface; and an ID decoding circuit, coupled to the first ID transmission path or the second ID transmission path, wherein the ID decoding circuit is configured to receive an ID signal and decode the ID signal to generate a data selection signal, wherein the semiconductor chip outputs data signals transmitted in a plurality of data transmission paths to corresponding internal receiving circuits respectively according to the data selection signal.

2. The semiconductor chip according to claim 1, wherein each of the internal receiving circuits comprises: a first data receiving circuit and a second data receiving circuit, wherein the first data receiving circuit is coupled to a first data selecting circuit, and the second data receiving circuit is

coupled to a second data selecting circuit; and the plurality of data transmission paths comprise: a first data transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, wherein the first data transmission path is configured to transmit a first data signal; a second data transmission path, penetrating through the substrate in the direction perpendicular to the active surface of the semiconductor chip, and symmetrically distributed with the first data transmission path based on the central axis, wherein the second data transmission path is configured to transmit a second data signal, and the second data signal is different from the first data signal; the first data selecting circuit, separately coupled to the ID decoding circuit, the first data transmission path, and the second data transmission path, wherein the first data selecting circuit is configured to output one of the first data signal or the second data signal to the first data receiving circuit according to the data selection signal; and the second data selecting circuit, separately coupled to the ID decoding circuit, the first data transmission path, and the second data transmission path, wherein the second data selecting circuit is configured to output the other of the first data signal or the second data signal to the second data receiving circuit according to the data selection signal.

3. The semiconductor chip according to claim 1, wherein the first ID transmission path is configured to transmit a first ID signal; and the second ID transmission path is configured to transmit a second ID signal, wherein the second ID signal is different from the first ID signal.

4. The semiconductor chip according to claim 1, wherein the first ID transmission path comprises: a first connection pad located on an inactive face of the semiconductor chip, a first path located in the substrate, and a first interconnection structure located on the active surface of the semiconductor chip, wherein the first connection pad, the first path, and the first interconnection structure are sequentially connected; and the second ID transmission path comprises: a second connection pad located on the inactive face of the semiconductor chip, a second path located in the substrate, and a second interconnection structure located on the active surface of the semiconductor chip, wherein the second connection pad, the second path, and the second interconnection structure are sequentially connected.

5. The semiconductor chip according to claim 4, wherein the ID decoding circuit is coupled to the first interconnection structure or the second interconnection structure.

6. A semiconductor structure, comprising: a logic chip; and a stack structure, stacked on the logic chip, and comprising: 1st to 2N-th semiconductor chips stacked sequentially, wherein the 1st to 2N-th semiconductor chips are same memory chips; an inactive face of the 1st semiconductor chip is bonded to the logic chip, and an active surface of a 2i-th semiconductor chip is bonded to an active surface of a 2i-1-th semiconductor chip; and N is a positive integer, and i is a positive integer less than or equal to N; and wherein each of the semiconductor chips comprises: a first ID transmission path, a second ID transmission path, and an ID decoding circuit; the first ID transmission path and the second ID transmission path penetrate through a substrate of one of the semiconductor chips in a direction perpendicular to an active surface of the corresponding one of the semiconductor chips and are symmetrically distributed based on a central axis of the corresponding one of the semiconductor chips, and the central axis passes through a center of the corresponding one of the semiconductor chips and is parallel to the active surface; and the ID decoding circuit is coupled to the first ID transmission path or the second ID transmission path, the ID decoding circuit is configured to receive an ID signal, and the ID signal indicates that one of the semiconductor chips is the 2i-th semiconductor chip or the 2i-1-th semiconductor chip in the stack structure, wherein the first ID transmission path of the 2i-1-th semiconductor chip and the second ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a first ID transmission channel; and the second ID transmission path of the 2i-1-th semiconductor chip and the first ID transmission path of the 2i-th semiconductor chip are sequentially connected to form a second ID transmission channel.

7. The semiconductor structure according to claim 6, wherein the logic chip is configured to: output

a first ID signal to the ID decoding circuit of the 2i-1-th semiconductor chip through one of the first ID transmission channel and the second ID transmission channel; and output a second ID signal to the ID decoding circuit of the 2i-th semiconductor chip through the other of the first ID transmission channel and the second ID transmission channel, wherein the second ID signal is different from the first ID signal.

8. The semiconductor structure according to claim 7, wherein the ID decoding circuit is configured to: decode the first ID signal to generate a first data selection signal, or decode the second ID signal to generate a second data selection signal, wherein the second data selection signal is different from the first data selection signal.

9. The semiconductor structure according to claim 7, wherein each of the semiconductor chips further comprises: a first data receiving circuit, a second data receiving circuit, a first data transmission path, a second data transmission path, a first data selecting circuit, and a second data selecting circuit, wherein the first data transmission path and the second data transmission path penetrate through the substrate in a direction perpendicular to an active surface of one of the semiconductor chips and are symmetrically distributed based on the central axis; the first data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the first data receiving circuit; and the second data selecting circuit is separately coupled to the ID decoding circuit, the first data transmission path, the second data transmission path, and the second data receiving circuit; and wherein the first data transmission path of the 2i-1-th semiconductor chip and the second data transmission path of the 2i-th semiconductor chip are sequentially connected to form a first data transmission channel; the second data transmission path of the 2i-1-th semiconductor chip and the first data transmission path of the 2i-th semiconductor chip are sequentially connected to form a second data transmission channel; the first data transmission channel is configured to transmit a first data signal; the second data transmission channel is configured to transmit a second data signal; and the second data signal is different from the first data signal.

10. The semiconductor structure according to claim 9, wherein the first data selecting circuit of the 2i-1-th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the 2i-1-th semiconductor chip according to the first data selection signal; the second data selecting circuit of the 2i-1-th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the 2i-1-th semiconductor chip according to the first data selection signal; the first data selecting circuit of the 2i-th semiconductor chip is configured to: output one of the first data signal and the second data signal to the first data receiving circuit of the 2i-th semiconductor chip according to the second data selection signal; and the second data selecting circuit of the 2i-th semiconductor chip is configured to: output the other of the first data signal and the second data signal to the second data receiving circuit of the 2i-th semiconductor chip according to the second data selection signal.

11. The semiconductor structure according to claim 6, wherein central axes of any two adjacent semiconductor chips among the 1st to 2N-th semiconductor chips are aligned, wherein the first ID transmission path of one of the two adjacent semiconductor chips is aligned with the second ID transmission path of the other of the two adjacent semiconductor chips.

12. The semiconductor structure according to claim 6, wherein an inactive face of a 2j-th semiconductor chip is bonded to an inactive face of a 2j+1-th semiconductor chip, wherein j is an integer greater than or equal to one, and 2j+1 is less than 2N.

13. The semiconductor structure according to claim 6, wherein the first ID transmission path comprises: a first connection pad located on an inactive face of one of the semiconductor chips, a first path located in the substrate, and a first interconnection structure located on an active surface of the corresponding one of the semiconductor chips, wherein the first connection pad, the first path, and the first interconnection structure are sequentially connected; and the second ID

transmission path comprises: a second connection pad located on an inactive face of one of the semiconductor chips, a second path located in the substrate, and a second interconnection structure located on an active surface of the corresponding one of the semiconductor chips, wherein the second connection pad, the second path, and the second interconnection structure are sequentially connected.

14. A semiconductor device, comprising: a substrate, wherein the semiconductor chip according to claim 1 is located on the substrate, and the semiconductor chip is bonded to the substrate; or, the semiconductor structure is located on the substrate, and the semiconductor structure is bonded to the substrate.

15. The semiconductor device according to claim 14, comprising: a high bandwidth memory.
